

Excel

Complete Notes



Excel Basics



1 What is Excel?



Excel is a powerful **spreadsheet** application by Microsoft used to store, organize, analyze and visualize data efficiently.

2 Workbook vs Worksheet

- **Workbook**: The entire Excel file with .xlsx extension.
- **Worksheet**: An individual sheet inside a workbook.

3 Rows, Columns & Cells

- **Rows**: Horizontal collection of cells. (Numbered 1, 2, 3 ...)
- **Columns**: Vertical collection of cells. (Labeled A, B, C ...)
- **Cells**: Intersection of a row and a column.

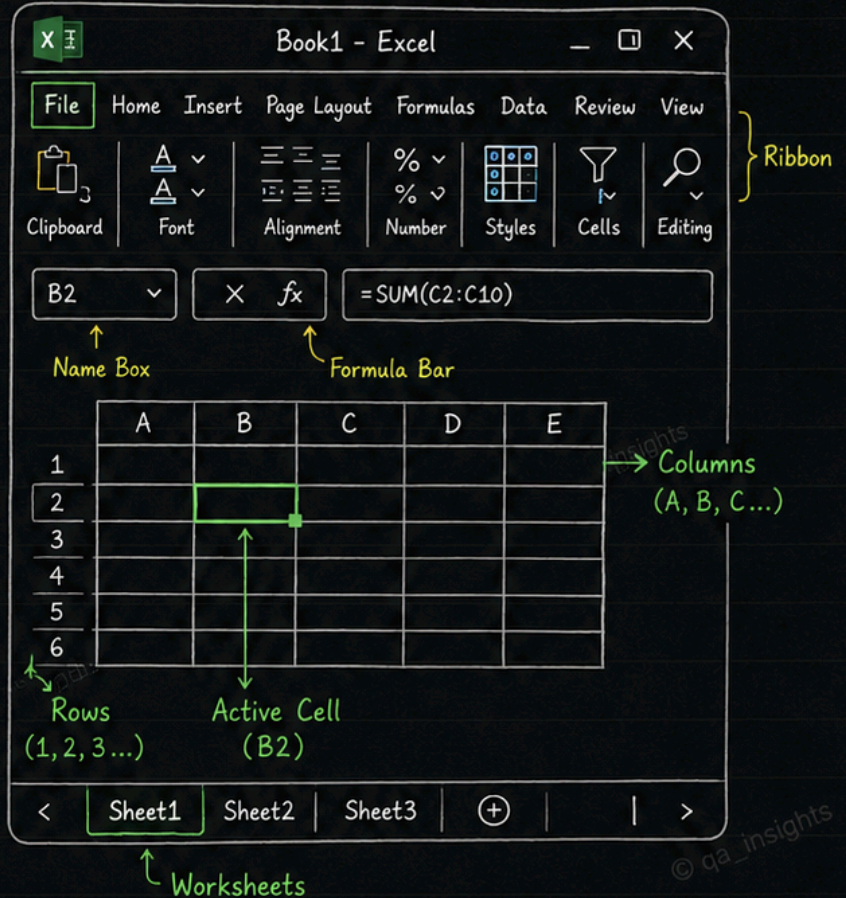


Example

	A	B	C	D
1				
2				
3				

Cell B2 is the intersection of Column B and Row 2.

4 Excel Interface Overview



Cell Address: Unique location of a cell (e.g., B2)



Name Box: Shows the address or name of the active cell.



Formula Bar: Displays and edits the content or formula of the active cell.



Ribbon: Contains tabs and commands to perform actions.



Best Practice

- Keep data clean and structured.
- Use meaningful sheet names.
- Use formulas instead of hardcoding values.



Interview Tip

- Know the difference between Workbook and Worksheet.
- Understand cell referencing (Relative: A1, Absolute: \$A\$1).
- Practice navigation using Ribbon and shortcuts.

Basic Formulas



1 Common Basic Formulas

- **SUM(range)** → Adds all numbers in the range.
- **AVERAGE(range)** → Returns the average value.
- **MIN(range)** → Returns the smallest value.
- **MAX(range)** → Returns the largest value.
- **COUNT(range)** → Counts cells with numbers.
- **COUNTA(range)** → Counts non-empty cells.

2 Example

	A	B
1	Item	Amount
2	Book	250
3	Pen	50
4	Bag	700
5	Total	1000

= SUM(B2:B4) → 1000
 = AVERAGE(B2:B4) → 333.33
 = MIN(B2:B4) → 50
 = MAX(B2:B4) → 700
 = COUNT(B2:B4) → 3
 = COUNTA(B2:B4) → 3



SUM, AVERAGE, MIN, MAX work only on numeric values. COUNTA counts any non-empty cell (text, number, date etc.).

3 Cell References in Excel

- **Relative Reference:** Changes when copied to another cell.
- **Absolute Reference:** Fixed reference, does not change on copy.
- **Mixed Reference:** Only row or column is fixed.

★ Best Practice

- ✓ Use meaningful range names.
- ✓ Prefer formulas over hardcoding values.
- ✓ Use absolute references in reports & dashboards.
- ✓ Double-check ranges before applying formulas.

Type	Example	How it behaves when copied
Relative (Changes) 	A1	If A1 is copied to B2, it becomes B2.
Absolute (Fixed) 	\$A\$1	If \$A\$1 is copied to B2, it remains \$A\$1.
Mixed (Column Fixed) 	\$A1	If \$A1 is copied to B2, it becomes \$A2.
Mixed (Row Fixed) 	A\$1	If A\$1 is copied to B2, it becomes B\$1.



Interview Tip

- Q. What is the difference between COUNT and COUNTA?
- A. COUNT counts only cells with numbers, while COUNTA counts all non-empty cells (numbers, text, dates, logical values, etc.).

Logical Functions



① What are Logical Functions?

- Logical functions in Excel help us make decisions, test conditions and return results based on **TRUE** or **FALSE**.



These functions are widely used in reports, scorecards, dashboards, validations and data quality checks.

② Common Logical Functions

Function	What it does	Syntax	Example	Result
IF()	Returns one value if a condition is TRUE and another if FALSE.	=IF(condition, value_if_true, value_if_false)	=IF(B2>=50, "Pass", "Fail")	Pass
IFS()	Checks multiple conditions and returns value for the first TRUE condition.	=IFS(condition1, value1, condition2, value2, ...)	=IFS(B2>=90, "A", B2>=75, "B", B2>=50, "C")	A / B / C
IFERROR()	Returns a custom value if a formula returns an error.	=IFERROR(value, value_if_error)	=IFERROR(A2/B2, "Cannot Divide")	Cannot Divide
AND()	Returns TRUE if all conditions are TRUE.	=AND(condition1, condition2, ...)	=AND(B2>50, C2<100)	TRUE / FALSE
OR()	Returns TRUE if any one condition is TRUE.	=OR(condition1, condition2, ...)	=OR(B2>50, C2<100)	TRUE / FALSE
NOT()	Reverses the logical value.	=NOT(logical)	=NOT(B2>50)	TRUE / FALSE

③ Nested IF

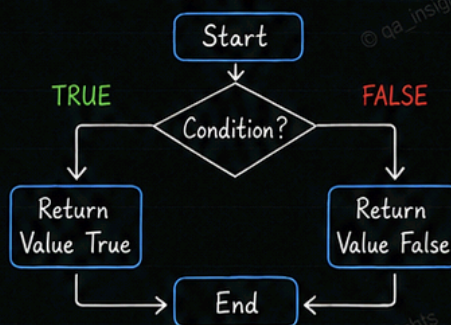
- An IF inside another IF to handle multiple conditions.

Example

=IF(B2>=90, "A", IF(B2>=75, "B", IF(B2>=50, "C", "Fail")))

} Checks multiple conditions step-by-step.

④ Quick Flow



☆ Best Practice

- ✓ Keep conditions simple.
- ✓ Use IFS() instead of multiple nested IFs.
- ✓ Use IFERROR() to handle errors gracefully.
- ✓ Combine AND(), OR(), NOT() for powerful validations.



Interview Tip

Q. Difference between IF and IFS?

A. **IF** checks one condition, while **IFS** checks multiple conditions in a clean and readable way.



Lookup & Reference



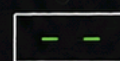
① What are Lookup Functions?

Lookup functions find and return data from a table or range based on a matching value.

Lookup



Return

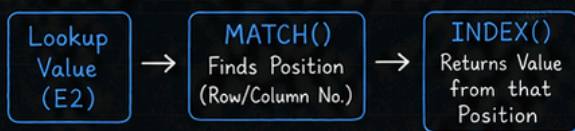


Result

② Common Lookup Functions

Function	What it does	Syntax (Basic)	Example	Notes
XLOOKUP() ★ (Recommended)	Searches a value in any direction and returns matching result.	=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found])	=XLOOKUP(D2, A2:A10, B2:B10, "Not Found")	✓ Works left to right or right to left ✓ Exact or Approx match ✓ Handles errors
VLOOKUP() 	Looks up a value in first column of a table and returns a value from the same row.	=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])	=VLOOKUP(D2, A2:C10, 3, FALSE)	⚠ Only searches left to right ⚠ Use FALSE for exact match
HLOOKUP() 	Looks up a value in top row of a table and returns a value from the same column.	=HLOOKUP(lookup_value, table_array, row_index_num, [range_lookup])	=HLOOKUP(D1, A1:Z5, 4, FALSE)	⚠ Searches top to bottom ⚠ Use FALSE for exact match
INDEX()	Returns the value of a cell in a given row and column.	=INDEX(array, row_num, column_num)	=INDEX(B2:D10, 3, 2)	✓ Very flexible ✓ Use with MATCH for powerful lookups
MATCH() 	Returns the position of a value in a row or column.	=MATCH(lookup_value, lookup_array, [match_type])	=MATCH(E2, A2:A10, 0) (Returns position)	✓ 0 = Exact match ✓ 1 = Less than ✓ -1 = Greater than
XMATCH() ★ (Enhanced)	Improved MATCH with more features and search options.	=XMATCH(lookup_value, lookup_array, [match_mode], [search_mode])	=XMATCH(E2, A2:A10, 0, -1)	✓ Searches in any direction ✓ More match modes ✓ Handles errors

③ How INDEX + MATCH Works



Powerful combination for flexible lookups in any direction! 💡

✓ Best Practice

- ✓ Prefer XLOOKUP for new formulas.
- ✓ Use INDEX + MATCH for maximum flexibility.
- ✓ Always use exact match (FALSE or 0) unless approximate match is needed.
- ✓ Keep lookup ranges clean and sorted (if required).

Example Table

ID	Name	Product	Sales
101	Ravi	Laptop	50000
102	Neha	Mobile	30000
103	Amit	Tablet	20000
104	Kiran	Watch	15000

Get Sales for ID 103 using INDEX+MATCH
 =INDEX(D2:D5, MATCH(103, A2:A5, 0)) → 20000

🏆 Interview Tip

- Q. Which is better: VLOOKUP or XLOOKUP?
- A. **XLOOKUP** is more powerful. It can search in both directions, handle errors, and return exact or approximate matches easily.

Text Functions



1 What are Text Functions?

- Text functions in Excel are used to extract, clean, combine and manipulate text data.



Why Important?

- Clean messy data
- Extract specific text
- Standardize text formats
- Useful in reports & dashboards

2 Common Text Functions

Function	What it does	Syntax	Example	Result
LEFT()	Returns the leftmost characters from a text.	=LEFT(text, num_chars)	=LEFT("QA Insights", 2)	QA
RIGHT()	Returns the rightmost characters from a text.	=RIGHT(text, num_chars)	=RIGHT("QA Insights", 8)	Insights
MID()	Returns characters from the middle of a text.	=MID(text, start_num, num_chars)	=MID("QA Insights", 4, 7)	Insights
LEN()	Returns the total number of characters in a text.	=LEN(text)	=LEN("QA Insights")	11
TRIM()	Removes extra spaces from text.	=TRIM(text)	=TRIM(" QA Insights ")	QA Insights
CONCAT()	Joins multiple text strings into one text.	=CONCAT(text1, text2, ...)	=CONCAT("QA", " ", "Insights")	QA Insights
TEXTJOIN()	Joins text with a delimiter and ignores empty cells.	=TEXTJOIN(delimiter, ignore_empty, text1, [text2], ...)	=TEXTJOIN(", ", TRUE, "QA", " ", "Insights")	QA, Insights
TEXTSPLIT()	Splits text into rows or columns by delimiter.	=TEXTSPLIT(text, col_delimiter, [row_delimiter], [ignore_empty])	=TEXTSPLIT("QA-Insights-2026", "-")	QA Insights 2026

3 Example - Full Usage

	A (Original Text)	B (Formula)	C (Result)
1	"QA Insights "	=TRIM(A2)	QA Insights
2	"QA Insights	=LEFT(A3,2)	QA
3	"QA Insights	=RIGHT(A3,8)	Insights
4	"QA Insights	=MID(A3,4,7)	Insights
5	"QA Insights	=LEN(A3)	11
6	"QA Insights	=CONCAT(A3, " - ", "Excel")	QA Insights - Excel
7	QA Insights 2026	=TEXTSPLIT(A8, " ")	QA Insights 2026



Best Practice

- ✓ Use TRIM() before matching or comparing data.
- ✓ Prefer TEXTJOIN() over CONCAT() for ranges.
- ✓ Use TEXTSPLIT() to break data into rows/columns quickly.
- ✓ Combine with other functions like IF(), FILTER(), UNIQUE().



Interview Tip

- Q. Difference between CONCAT() and TEXTJOIN()?
- A. CONCAT() joins text but includes empty cells. TEXTJOIN() allows a delimiter and can ignore empty cells using the ignore_empty argument (TRUE).



Real World Use

- Data cleaning & standardization
- Preparing reports & dashboards
- Parsing logs, emails, IDs, addresses

Date & Time Functions



1 What are Date & Time Functions?

- These functions help us work with dates & time values.
- Useful for calculations, reporting, scheduling and SLA tracking.



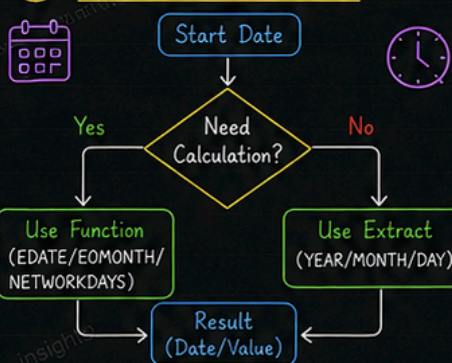
Why Important?

- Automate date based calculations
- Handle deadlines, durations & aging
- Improve accuracy in reports & dashboards
- Essential in real world projects

2 Common Date & Time Functions

Function	What it does	Syntax (Basic)	Example	Result
TODAY()	Returns today's date.	=TODAY()	=TODAY()	15-May-2026
NOW()	Returns current date and time.	=NOW()	=NOW()	15-May-2026 10:30 AM
DATE()	Creates a date from year, month, day.	=DATE(year, month, day)	=DATE(2026, 5, 15)	15-May-2026
YEAR()	Extracts the year from a date.	=YEAR(date)	=YEAR(A2)	2026
MONTH()	Extracts the month from a date.	=MONTH(date)	=MONTH(A2)	5
DAY()	Extracts the day from a date.	=DAY(date)	=DAY(A2)	15
EDATE()	Returns the date that is N months before or after start date.	=EDATE(start_date, months)	=EDATE(A2, 3)	15-Aug-2026
EOMONTH()	Returns the last day of the month N months before or after start date.	=EOMONTH(start_date, months)	=EOMONTH(A2, 0)	31-May-2026
NETWORKDAYS()	Returns number of workdays between two dates.	=NETWORKDAYS(start_date, end_date, [holidays])	=NETWORKDAYS(A2, B2, C2:C10)	18

3 Quick Reference Flow



Example Box

Employee Joining Date in A2
= 15-Feb-2024

- Completion after 6 months
=EDATE(A2, 6) → 15-Aug-2024
- Last day of that month
=EOMONTH(A2, 6) → 31-Aug-2024
- Days worked till 15-May-2026
=NETWORKDAYS(A2, TODAY())
→ 587
- Year of Joining
=YEAR(A2) → 2024

★ Best Practice

- ✓ Store dates as real date values, not as text.
- ✓ Use TODAY() instead of typing dates (dynamic).
- ✓ Use NETWORKDAYS() for SLA & business days calculations.
- ✓ Combine with IF(), DATEDIF(), WORKDAY() for advanced use.



Interview Tip

Q. What is the difference between EDATE() and EOMONTH()?

A. EDATE() returns the same day in future/past month, while EOMONTH() returns the last day of that month.

Data Cleaning & Validation



① What is Data Cleaning?

- **Data Cleaning** is the process of fixing or removing incorrect, incomplete, duplicate or inconsistent data.
- Clean data leads to accurate reports, better decisions & reliable analysis.



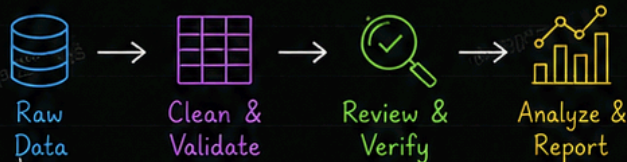
Why Important?

- Improves data accuracy & quality
- Saves time in reporting & analysis
- Prevents wrong business decisions
- Essential for dashboards, KPIs & SLA tracking

② Key Data Cleaning & Validation Features

Feature	What it does	How it helps	Example / Use Case
Sort	Arranges data in Ascending or Descending order.	Easier to analyze large datasets.	Sort employees by Salary (High to Low).
Filter	Displays only rows that meet criteria.	Focus on specific records quickly.	Filter only "Passed" test cases.
Remove Duplicates	Removes duplicate rows from data.	Keeps dataset unique & clean.	Remove duplicate Employee IDs.
Flash Fill	Automatically fills data based on pattern.	Saves time on repetitive entries.	Split First & Last Name automatically.
Text to Columns	Splits text in a cell into multiple columns.	Converts unstructured data into columns.	Split "City,State,Country" into 3 columns.
Data Validation	Restricts input based on rules/criteria.	Prevents invalid or incorrect data entry.	Allow only "Pass/Fail" in Result column.
Conditional Formatting	Highlights cells based on conditions.	Visual indicators for quick insights.	Highlight due dates that are past due.

③ Workflow



☆ Good data in → Good insights out!

④ Best Practices

- ✓ Always validate data at the point of entry.
- ✓ Remove duplicates before analysis.
- ✓ Use clear rules in Data Validation.
- ✓ Use Conditional Formatting for exceptions.
- ✓ Keep source data safe, work on a copy.



Interview Tip

Q. Why is data cleaning important in Excel?

A. Clean data ensures accuracy, reduces errors, and helps in reliable reporting & decision making.

Pivot Tables & Charts



① What are Pivot Tables?

- **Pivot Tables** are powerful tools in Excel used to **summarize, analyze, explore** and present large datasets quickly.
- Convert raw data into **meaningful insights** without writing complex formulas.
- Help in **reporting, trend analysis, comparisons** and decision making.



Why Important?

- Summarize thousands of rows in seconds
- Easy to group, filter and drill down data
- Interactive & dynamic reports
- Supports dashboards and business intelligence
- Essential for real world projects & analytics

② Key Features

Feature	What it does	Example / Use Case	Benefit
Pivot Table	Summarizes data by rows, columns and values.	Total Sales by Region, Product and Month.	Quick summaries & deep insights.
Pivot Chart	Creates charts from pivot table data.	Column chart for Sales trend by Month.	Visualize data for better understanding.
Slicers	Add buttons to filter pivot data interactively.	Filter data by Region, Category or Status.	User friendly & fast data filtering.
Timelines	Filter date fields using a timeline slider.	Filter data by Year, Quarter or Months.	Easy time based analysis.
Recommended Charts	Suggests best chart type for your data.	Excel recommends chart based on data pattern.	Save time & choose right visual.
Dashboard Basics	Combine pivot tables, charts, slicers & KPIs in one view.	Sales Dashboard with KPIs, trends, slicers & charts.	Complete business overview in one page.

③ How it Works (Simple Flow)



☆ Refresh data → Pivot updates → Dashboard stays dynamic!

④ Recommended Charts Guide

Data Type / Goal	Best Chart
Compare categories	Column Chart
Show trend over time	Line Chart
Show part to whole	Pie / Doughnut
Compare multiple items	Bar Chart
Show relationship	Scatter Chart
Show progress to goal	Gauge / KPI



Tip
Let Excel suggest charts first, then refine based on your story and audience.

✓ Best Practices

- ✓ Keep source data clean before creating pivot.
- ✓ Use meaningful field names.
- ✓ Refresh pivot whenever source data changes.
- ✓ Use slicers & timelines for interactivity.
- ✓ Keep dashboard simple, clear & focused.



Interview Tip

- Q. What is the difference between Pivot Table and Table in Excel?
- A. Table is used to store and manage data with formatting and filters. **Pivot Table** is used to summarize, analyze and report data dynamically.

Advanced Excel



1 Key Advanced Functions

Function	What it does	Syntax (Basic)	Example	Result / Use Case
FILTER()	Returns filtered records based on condition.	=FILTER(array, include, [if_empty])	=FILTER(A2:D100, B2:B100="Passed", "No Data")	All passed test cases.
SORT()	Sorts data by one or more columns.	=SORT(array, sort_index, [sort_order])	=SORT(A2:D100, 3, -1)	Sort by 3rd column (descending).
UNIQUE()	Returns unique values from a range.	=UNIQUE(array, [by_col], [exactly_once])	=UNIQUE(B2:B100)	Unique values list.
SEQUENCE()	Generates sequential numbers, dates or arrays.	=SEQUENCE(rows, [columns], [start], [step])	=SEQUENCE(5,3,1,1)	Dynamic list 1 to 15 (5 rows x 3 cols).
LET()	Assigns names to calculations for reuse & readability.	=LET(name1, calc1, name2, calc2, ..., result)	=LET(Total, SUM(C2:C100), Tax, Total*0.18, Total+Tax)	Easier formulas, better performance.
LAMBDA()	Creates custom reusable functions.	=LAMBDA(params, calculation)	=LAMBDA(x, x*0.18)	Build your own Excel functions.
Dynamic Arrays	Spills results automatically into ranges.	Most modern functions return dynamic arrays (FILTER, SORT, UNIQUE...)	=FILTER(A2:D100, B2:B100="Open")	Results spill automatically.
Named Ranges	Create names for cells or ranges to use in formulas easily.	=NameManager or use name in formula	SalesRange = C2:C100 =SUM(SalesRange)	Improves readability & maintainability.

2 Dynamic Arrays in Action

Data (A1:D5)

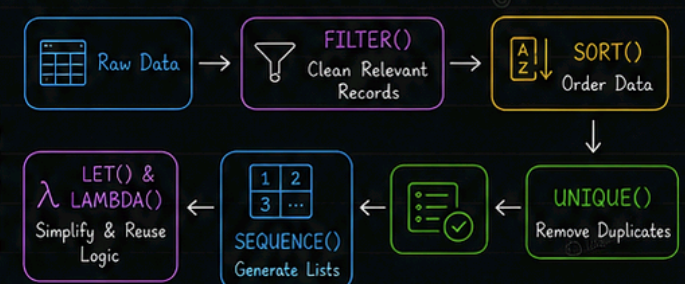
ID	Status	Owner	Priority
1	Open	Ravi	High
2	Closed	Neha	Low
3	Open	Amit	Medium
4	Open	Kiran	High
5	Closed	Priya	Low

Formula in F2
=FILTER(A2:D5, B2:B5="Open")

Result (Spilled)

ID	Status	Owner	Priority
1	Open	Ravi	High
3	Open	Amit	Medium
4	Open	Kiran	High

3 How They Work Together



Best Practices

- ✓ Prefer **FILTER()** over complex IF() + copy-paste solutions.
- ✓ Use **LET()** to break complex formulas into readable steps.
- ✓ Use **Named Ranges** for important datasets and parameters.
- ✓ Combine functions (**FILTER + SORT + UNIQUE**) for powerful reports.
- ✓ Keep formulas dynamic to reduce manual updates.

Example Box

Calculate Total Sales After Discount

```
=LET(
  Sales, C2:C100,
  Discount, D2:D100,
  Total, Sales*(1-Discount),
  SUM(Total))
```

Interview Tip

- Q. What is the advantage of Dynamic Arrays?
- A. They automatically spill results into adjacent cells, reducing manual work and making reports more flexible.

Automation & Productivity



1 Automation Features in Excel

Feature	What it does	Why it's useful	Example / Use Case
Power Query	Import, clean, transform and combine data from multiple sources.	Automates data preparation & refresh.	Load daily test results from CSV, clean & generate reports.
Power Pivot	Create data model, relationships & DAX measures.	Handle large datasets and advanced calculations.	Build data model for test metrics (Defects, Execution, Coverage).
Macros	Record actions and replay them automatically.	Saves time on repetitive tasks.	Format reports, apply filters, export data.
VBA Basics	Write custom scripts (Visual Basic for Applications).	Automate complex logic and user defined functions.	Custom validation, auto email reports, buttons.
Freeze Panes	Keep rows/columns visible while scrolling.	Improves readability of large datasets.	Lock header row while reviewing long test result sheets.
Protect Sheet	Lock cells/structure and restrict changes.	Prevents accidental edits, maintains integrity.	Protect formulas and critical data in reports.

2 Excel for QA & Data Analysis

- Analyze test execution, defects, requirements and coverage.
- Create dashboards, trend reports & KPIs.
- Validate data quality and business rules.
- Track metrics like Pass %, Defect Density, Reopen Rate.
- Integrate with APIs, Databases and Automation tools.



3 Keyboard Shortcuts (Must Know)

Ctrl + T	→ Create Table
Ctrl + Shift + L	→ Toggle Filter
Ctrl + Arrow Keys	→ Jump to edge of data
Ctrl + Shift + \$	→ Apply Currency format
Ctrl + 1	→ Format Cells
Alt + =	→ AutoSum
Ctrl + Y	→ Redo
Ctrl + Z	→ Undo
Ctrl + S	→ Save
F4	→ Repeat last action



Productivity Tip

Automate the boring, focus on testing & analysis!



4 Example - QA Dashboard Flow



Import Test Data (Power Query)



Model Data & Create Measures (Power Pivot)



Analyze & Visualize (Charts/Pivots)



Dashboard & Reports

Clean Data → Model → Analyze → Visualize → Decide

Best Practices

- ✓ Use Power Query for repeatable data preparation.
- ✓ Prefer Power Pivot + DAX for large scale analysis.
- ✓ Record Macros for simple tasks, use VBA for complex logic.
- ✓ Protect important sheets and cells.
- ✓ Use shortcuts to reduce manual effort.
- ✓ Keep dashboards clean, interactive and meaningful.

Interview Tip

Q. When would you use Power Query over formulas?

A. Use Power Query when:

- Data source changes frequently.
- Large data needs cleaning/transformations.
- Multiple steps are required before analysis.

It is faster, repeatable and reduces manual errors.

Excel Formula Cheat Sheet (Part 1)



Math & Statistics

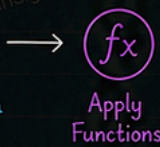


#	Formula	What it does	Syntax (Basic)	Example	Result
1	SUM() Σ	Adds all numbers in a range.	=SUM(number1, [number2], ...)	=SUM(B2:B10)	1250
2	AVERAGE() $\bar{x}$	Returns the average (Arithmetic mean).	=AVERAGE(number1, [number2], ...)	=AVERAGE(B2:B10)	125
3	COUNT() #	Counts cells that contain numbers.	=COUNT(value1, [value2], ...)	=COUNT(B2:B10)	10
4	COUNTA() $\equiv$	Counts cells that are not empty.	=COUNTA(value1, [value2], ...)	=COUNTA(A2:A10)	10
5	COUNTIF() $\checkmark$	Counts cells that meet a single criteria.	=COUNTIF(range, criteria)	=COUNTIF(B2:B10, ">100")	6
6	COUNTIFS() $\checkmark\checkmark$	Counts cells that meet multiple criteria.	=COUNTIFS(range1, criteria1, [range2, criteria2], ...)	=COUNTIFS(B2:B10, ">100", C2:C10, "East")	3
7	SUMIF() $\Sigma\checkmark$	Adds cells that meet a single criteria.	=SUMIF(range, criteria, [sum_range])	=SUMIF(C2:C10, "East", B2:B10)	780
8	SUMIFS() $\Sigma\checkmark\checkmark$	Adds cells that meet multiple criteria.	=SUMIFS(sum_range, range1, criteria1, [range2, criteria2], ...)	=SUMIFS(B2:B10, C2:C10, "East", D2:D10, ">=2025-01-01")	950
9	AVERAGEIF() $\bar{x}\checkmark$	Returns average for cells that meet criteria.	=AVERAGEIF(range, criteria, [average_range])	=AVERAGEIF(C2:C10, "West", B2:B10)	142.5
10	ROUND() $\approx$	Rounds a number to a specified digits.	=ROUND(number, num_digits)	=ROUND(123.456, 2)	123.46
11	ROUNDUP() $\uparrow$	Rounds a number away from zero.	=ROUNDUP(number, num_digits)	=ROUNDUP(123.451, 2)	123.46
12	ROUNDDOWN() $\downarrow$	Rounds a number toward zero.	=ROUNDDOWN(number, num_digits)	=ROUNDDOWN(123.459, 2)	123.45
13	ABS()	Returns absolute value (positive value).	=ABS(number)	=ABS(-45.67)	45.67
14	MOD() %	Returns the remainder after division.	=MOD(number, divisor)	=MOD(17, 5)	2

How They Work Together



Raw Data



Apply Functions



Summarize & Analyze



Insightful Reports



Clean Data + Right Formulas = Accurate Insights



Best Practices

- ✓ Use **SUMIFS/COUNTIFS** for multi-criteria analysis.
- ✓ Always lock ranges with \$ for stable formulas.
- ✓ Use **AVERAGEIF** instead of complex IF + AVERAGE.
- ✓ Round values only in final reports.
- ✓ Keep formulas simple, documented & tested.



Interview Tip

Q. What is the difference between **COUNT()** and **COUNTA()**?

A. **COUNT()** counts only cells with numbers, while **COUNTA()** counts non-empty cells (text, numbers, blanks with spaces, etc.).

Example Data (Range B2:B10)

120	80	150	200	75
100	130	90	110	95

Total (SUM) = 1250 | Average = 125

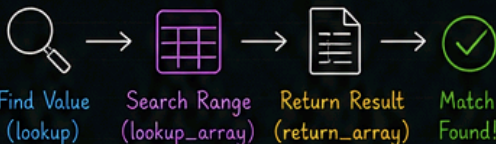
Excel Formula Cheat Sheet (Part 2)

Lookup & Text



#	Formula	What it does	Syntax (Basic)	Example	Result
1	XLOOKUP() 🔍	Looks up a value in a range and returns a matching result.	=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode], [search_mode])	=XLOOKUP(101, A2:A10, B2:B10, "Not Found")	Product A
2	VLOOKUP() 📊	Looks up a value in the first column and returns value from another column.	=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])	=VLOOKUP(101, A2:D10, 3, FALSE)	\$250
3	INDEX() 📊	Returns value of a cell in a given row & column of a range.	=INDEX(array, row_num, [column_num])	=INDEX(B2:D10, 3, 2)	\$450
4	MATCH() 🎯	Returns position of a value in a range.	=MATCH(lookup_value, lookup_array, [match_type])	=MATCH(101, A2:A10, 0)	2
5	IF() ⚡	Performs logical test and returns a value if TRUE or FALSE.	=IF(logical_test, value_if_true, value_if_false)	=IF(C2>=500, "Pass", "Fail")	Pass
6	IFERROR() ⚠️	Returns custom value if a formula results in error.	=IFERROR(value, value_if_error)	=IFERROR(VLOOKUP(999, A2:D10, 2, FALSE), "NA")	NA
7	LEFT() ←	Returns leftmost characters from text.	=LEFT(text, [num_chars])	=LEFT("QA Insights", 2)	QA
8	RIGHT() →	Returns rightmost characters from text.	=RIGHT(text, [num_chars])	=RIGHT("QA Insights", 8)	Insights
9	MID() []	Returns characters from middle of text.	=MID(text, start_num, [num_chars])	=MID("QA Insights", 4, 8)	Insights
10	LEN() 📏	Returns total number of characters in text.	=LEN(text)	=LEN("QA Insights")	11
11	TRIM() 🧹	Removes extra spaces from text.	=TRIM(text)	=TRIM(" QA Insights ")	QA Insights
12	TEXTJOIN() 🔗	Joins text from a range with a delimiter.	=TEXTJOIN(delimiter, ignore_empty, text1, [text2], ...)	=TEXTJOIN(", ", TRUE, A2:A4)	A, B, C
13	FILTER() ⚙️	Returns filtered records that meet condition.	=FILTER(array, include, [if_empty])	=FILTER(A2:C10, C2:C10="Pass")	Filtered Rows
14	UNIQUE() ★	Returns unique values from a range.	=UNIQUE(array, [by_col], [exactly_once])	=UNIQUE(A2:A10)	Unique List
15	SORT() ⬆️⬆️	Sorts data in ascending or descending order.	=SORT(array, [sort_index], [sort_order], [by_col])	=SORT(A2:C10, 2, -1)	Sorted Data

Lookup Flow



Best Practices

- ✓ Prefer **XLOOKUP** over **VLOOKUP** for flexible lookups.
- ✓ Use **IFERROR** to handle errors gracefully.
- ✓ Use **TRIM** & **CLEAN** before comparing text.
- ✓ Combine **FILTER** + **UNIQUE** + **SORT** for powerful reports.
- ✓ Keep formulas simple, readable & well documented.



Interview Tip

Q. What is the difference between **VLOOKUP** and **XLOOKUP**?

A.

- **VLOOKUP** can only look left to right, **XLOOKUP** can search both directions.
- **XLOOKUP** returns exact match by default and is easier to use.
- **XLOOKUP** handles missing data better with **if_not_found**.

Example Box

Data (A1:C5)

ID	Name	Score
101	Ravi	450
102	Neha	560
103	Amit	620
104	Priya	480

Example:

=XLOOKUP(103, A2:A5, B2:B5)

Result: Amit